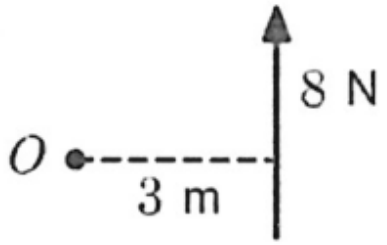
Moments

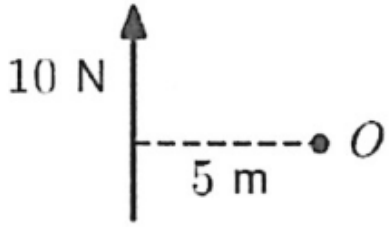
Questions 1–39

2.60 — Question 1. Find the resultant moment about O in each case, stating whether it is anticlockwise or clockwise:

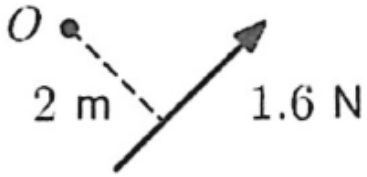
a)



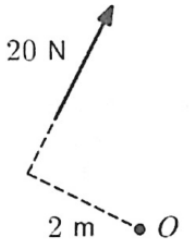
b)



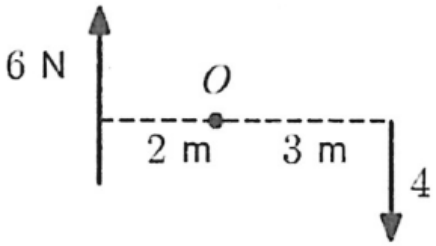
c)



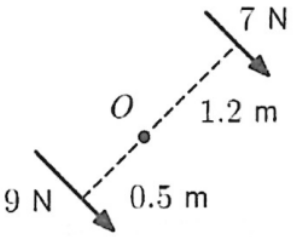
d)



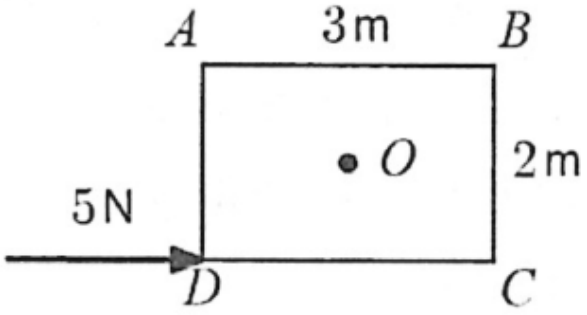
e)



f)



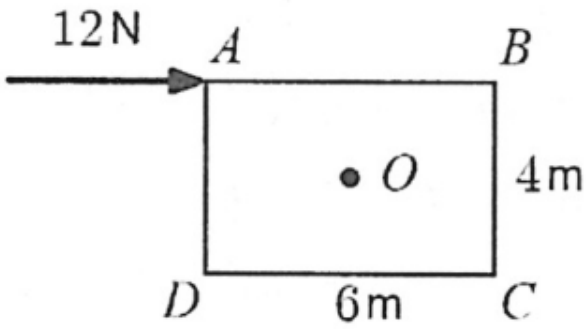
2.60 — Question 2. A uniform lamina $ABCD$ has height 2 metres and width 3 metres. It has a centre of mass at O which lies exactly where the lines AC and BD intersect. A force of 5 N acts on the point D parallel to DC as shown in the diagram.



Find the moment about

a) A
b) B
c) C
d) D
e) O

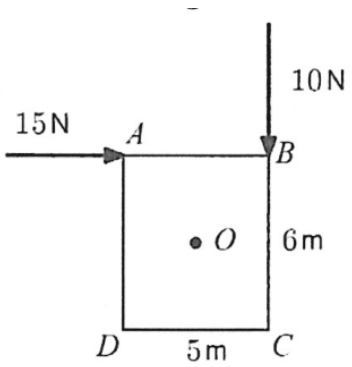
2.60 — Question 3. A uniform lamina $ABCD$ has height 4 metres and width 6 metres. The point O lies exactly where the lines AC and BD intersect. A force of 12 N acts on the point A parallel to AB as shown in the diagram.



Find the moment about

a) A
b) B
c) C
d) D
e) O

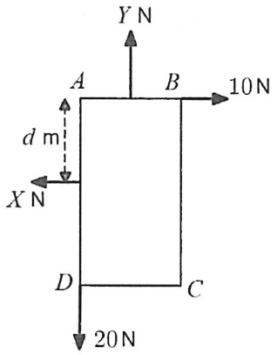
2.60 — Question 4. A uniform lamina $ABCD$ has height 6 metres and width 5 metres. The point O lies exactly where the lines AC and BD intersect. A force of 12 N acts on the point A parallel to AB and a force of 10 N acts on the points B parallel to BC as shown in the diagram.



Take moments about

a) A
b) B
c) C
d) D
e) O

2.60 — Question 5. A uniform rectangular lamina $ABCD$ is such that $BC = 0.2$ m and $AB = 0.08$ m. Forces of magnitude 10 N, 20 N, X N and Y N act on it as shown in the diagram, where the force of Y N acts at the midpoint of AB.



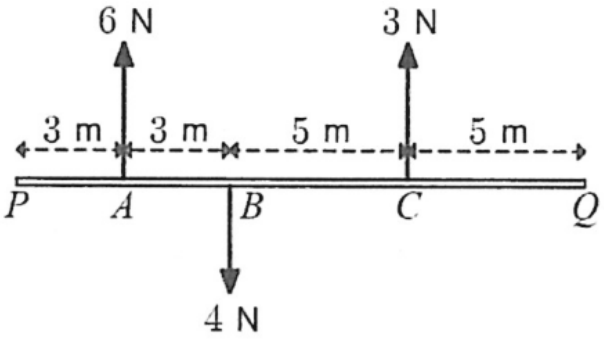
The lamina is in equilibrium. Find the value of

a) X
b) Y
c) d

The force acting at B is moved down towards point C.

d) Describe what happens to the lamina.

2.60 — Question 6. PQ is a light horizontal rod of length 16 m. Forces act on the rod as shown in the diagram

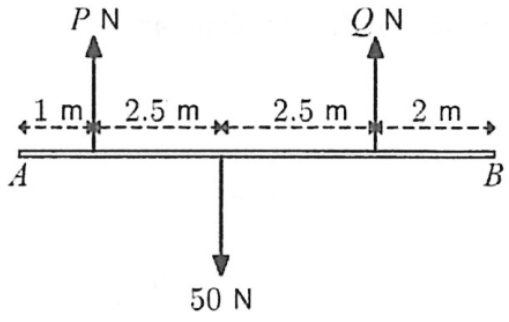


Find the resultant moment about

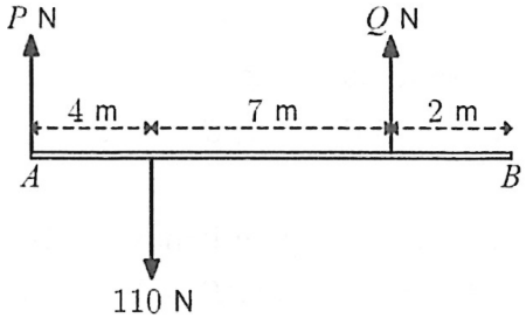
a) P
b) A
c) B
d) C
e) Q

2.60 — Question 7. Each of the diagrams show a horizontal beam AB which is in equilibrium. Two forces acting on the beam are given by P N and Q N.
In each case, find the values of P and Q:

a)

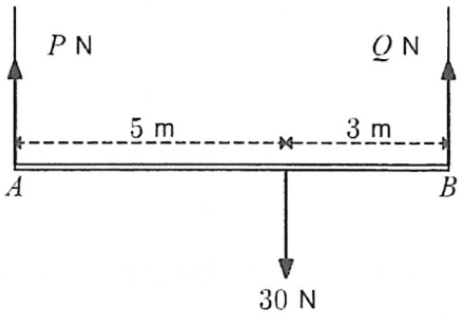


b)

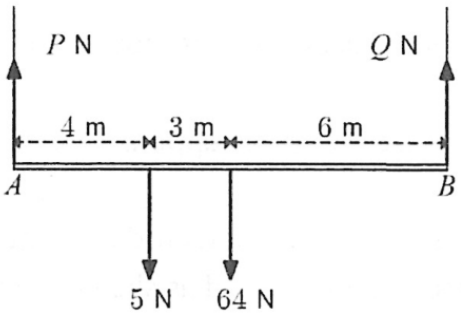


2.60 — Question 8. Each of the diagrams show a horizontal beam AB which is in equilibrium, being held in place by two inextensible cables, with tension P N and Q N respectively.
In each case, find the values of P and Q:

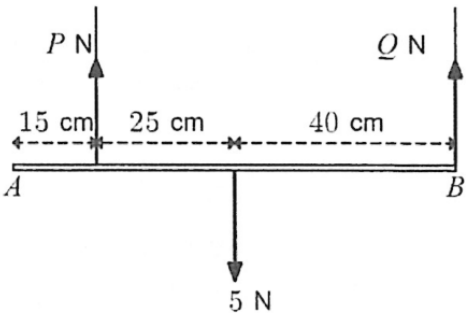
a)



b)

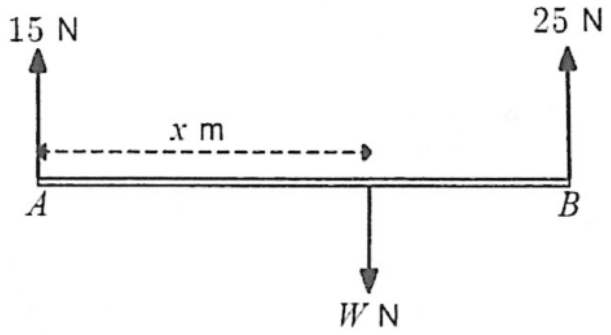


c)

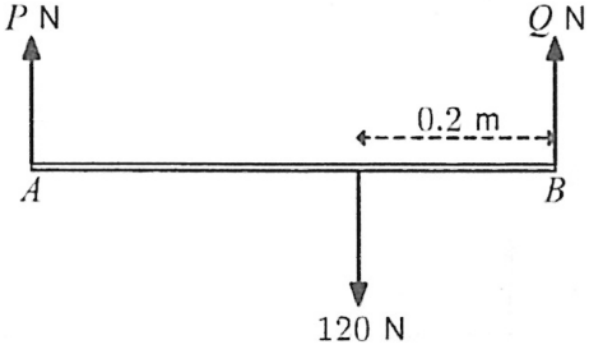


2.60 — Question 9. The diagrams below show a light horizontal rod PQ of length 0.5 m resting in equilibrium. Find

a) the value of x and W



b) the value of P and Q



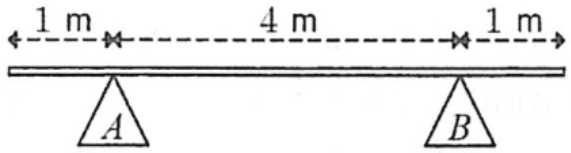
2.60 — Question 10. In each of the following cases, sketch a diagram, labelling the forces and distances:

a) 3 m uniform rod AB of weight 20 N, held up by two smooth supports at A and B.

b) 9 m uniform rod AB of weight 20 N, held by vertical ropes at A and B, with an object of weight 10 N which is 1m along from A.

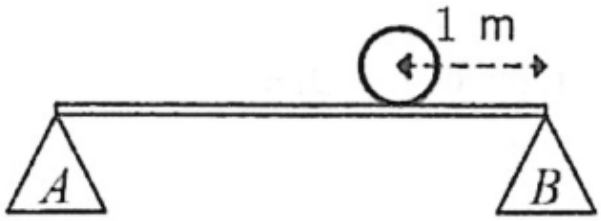
c) x m uniform rod AB of weight W N, held by vertical ropes at A and B, with an object of weight F N hanging from the rod d m along from A.

2.60 — Question 11. A uniform rod of weight 30 N rests horizontally on two smooth supports at A and B as shown.



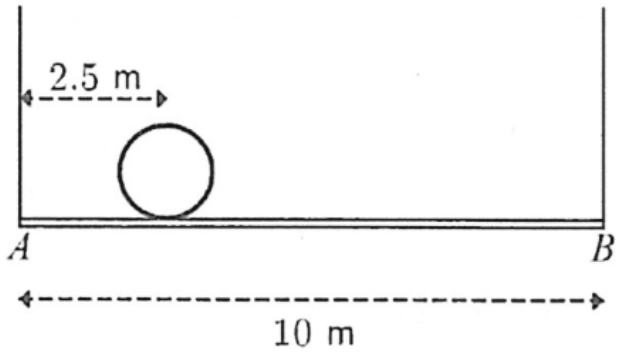
Find the reaction at A and the reaction at B.

2.60 — Question 12. A uniform rod AB of length 3 m and weight 60 N. It rests horizontally on two smooth supports at A and B. A particle of weight 12 N is positioned on the rod 1 metre from B



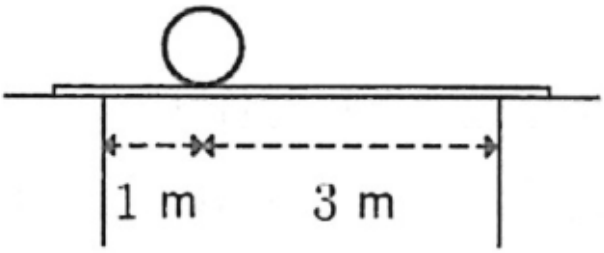
Find the reaction at A and the reaction at B.

2.60 — Question 13. The following questions use $g = 9.8$ **ms**$^{-2}$ A uniform rod AB of length 10 m and mass 5 kg is held in place by two inextensible cables at A and B respectively. A particle of mass 0.5 kg is positioned on the rod 2.5 metres from A



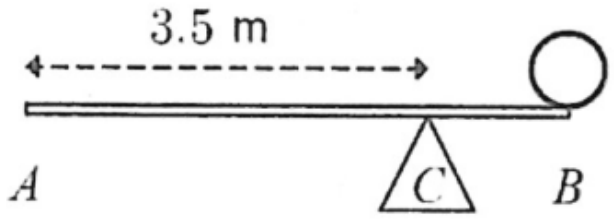
Given the rod is horizontal and in equilibrium, find the tension in the cables at A and B.

2.60 — Question 14. A plank of weight 500 N rests horizontally across two crates. An object of weight 200 N is placed 1 metre from the left-hand side as shown in the diagram.



a) Find the reaction forces exerted on each end of the plank. You may assume they act on the corners of the crates.
b) State the modelling assumptions you have made.

2.60 — Question 15. A uniform rod AB of length 4 m and mass 2 kg rests horizontally on a smooth support at C, which is 3.5 metres from A.
A particle of weight W N is positioned at B so that the rod is held in equilibrium.



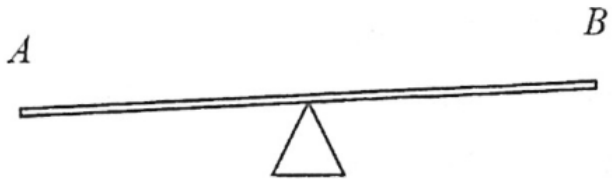
Find

a) the reaction at C.
b) the value of W
c) the mass of the particle.

An extra particle of weight W is added 0.5 metres to the left of C and the system remains in equilibrium.

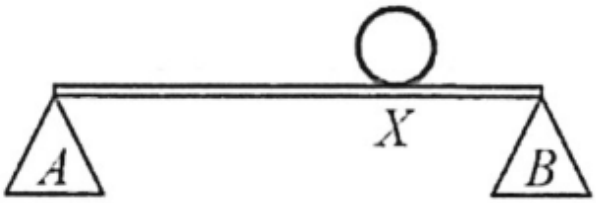
d) Find the new value of R

2.60 — Question 16. Alice and Bob are 22 kg and 18 kg respectively. They sit on a seesaw of length 5 metres with mass 40 kg, pivoted at its mid point. Bob sits at the right-hand side of the seesaw.



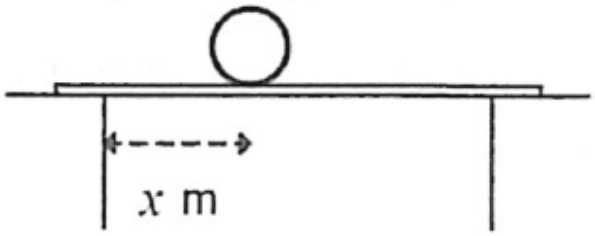
By modelling Alice and Bob as particles and the seesaw as a uniform rod, find how far Alice must sit from the left-hand end of the seesaw for the seesaw to be in equilibrium.

2.60 — Question 17. A uniform rod AB of length 0.5 m and mass 5 kg rests horizontally on two smooth supports at A and B as shown.



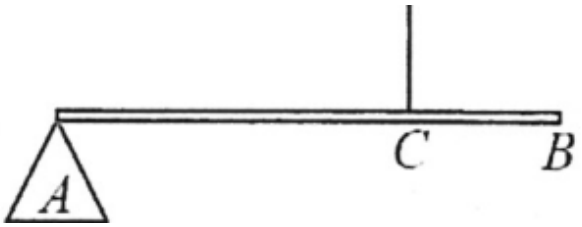
A particle of mass 8 kg is placed on the rod at X. The reaction on the rod at B is three times the reaction at A. Find the distance AX.

2.60 — Question 18. A plank of mass 10 kg rests horizontally on two crates which lie 8 metres apart. A particle of weight 1.5 kg is placed x metres from the left-hand side where $x < 4$ as shown.



One normal reaction is 20% larger than the other. By modelling the plank as a uniform rod, find the value of x

2.60 — Question 19. A uniform rod AB of weight 300 N is held horizontally in position by a smooth support at A and a vertical string at C.

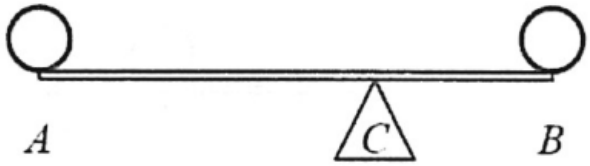


The string is 80% of the way along AB. Find the reaction at A and the tension in the string.

2.60 — Question 20. A non-uniform rod AB of length 3 metres and weight W N rests horizontally on two smooth supports at A and B. The reaction at A and B have magnitude 12 N and 18 N respectively.
Calculate the distance of the centre of mass of the rod from A.

2.60 — Question 21. A non-uniform rod AB of weight W newtons and length 10 metres rests horizontally on two smooth supports at A and B. The reaction at A and B have magnitude R N and 40 N respectively. Calculate the distance from A of the centre of mass of the rod giving your answer in terms of W

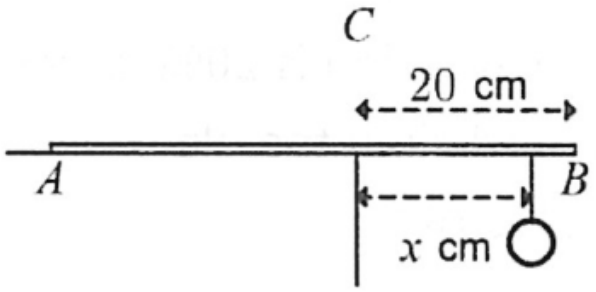
2.60 — Question 22. The following questions use exact values of g A uniform rod AB of length 5 metres and mass 5 kg. A particle of mass m kg is placed at A. A particle of mass $2m$ kg is placed at B. The rod is horizontal and at rest, balancing on a smooth support at C.



The distance between A and C is x metres.
Show that

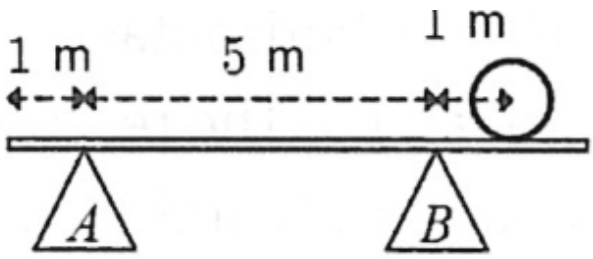
$$x = \frac{20m + 25}{6m + 10}$$

2.60 — Question 23. A uniform rod AB of mass 0.4 kg and length 0.7 m is resting on a table, with 20 cm of it overhanging the table. An object of mass m kg hangs from the rod x cm away from the table, as shown in the diagram, so that the rod is on the point of tilting.



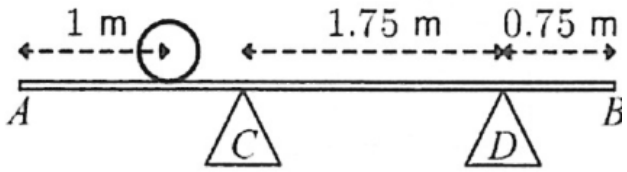
a) Show that $x = \frac{3}{50m}$
b) Find the range of values that m may be so that the rod remains horizontal.

2.60 — Question 24. A uniform rod of length 8 m and weight 60 N rests horizontally in equilibrium on two smooth supports at A and B as shown on the diagram. A particle of weight W N is positioned on the rod 1 m from B as shown. The rod is on the point of tilting.



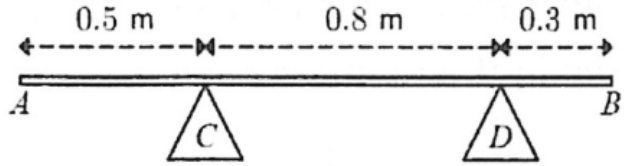
Find the reaction at B and the weight W

2.60 — Question 25. The following questions use $g = 10$ **ms**$^{-2}$ A uniform rod AB of length 4 m and mass 5 kg rests horizontally in equilibrium on two smooth supports at C and D as shown on the diagram. A particle of mass m kg is placed on the rod positioned on the rod 1 m from A as shown. The rod is on the point of tilting.



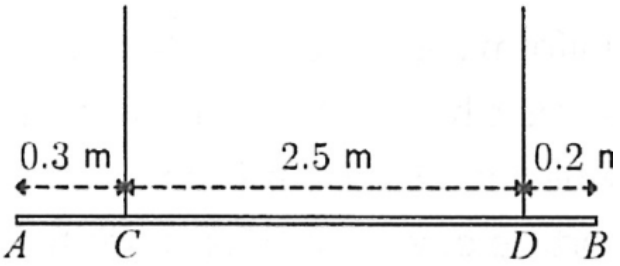
Find the reaction at C and the mass m

2.60 — Question 26. A uniform rod AB of length 1.6 m and weight W N rests horizontally in equilibrium on two smooth supports at C and D as shown on the diagram. A particle of mass $4W$ N is placed on the rod.



By taking moments about A, find the range of distances from A that the particle may be without it tipping.

2.60 — Question 27. A non-uniform plank AB of length 3 metres and mass 2 kg is held horizontally in equilibrium by two ropes that are positioned at C and D where $AC = 0.3$ m and $DB = 0.2$ m.



The tension in the rope at C is twice the tension in the rope at D.

a) Calculate the exact distance of the centre of mass from A.

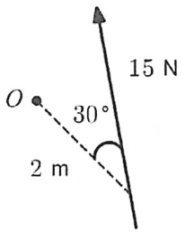
b) State the modelling assumptions you have used.

A child of mass 25 kg wants to walk across the plank. The ropes will break if either of the tensions exceed 200 N.

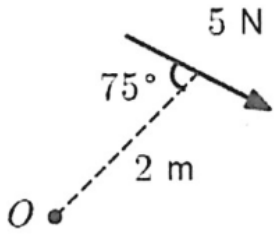
c) Find the start and end point where the child can walk without the ropes breaking, giving your answer to 2 significant figures.

2.61 — Question 1. Find the resultant moment about O in each case, giving your answer to 3 significant figures and stating whether it is anticlockwise or clockwise:

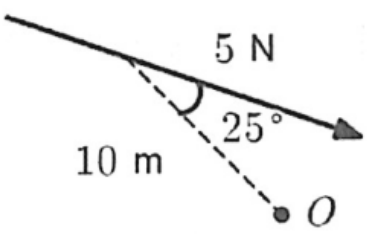
a)



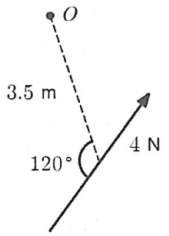
b)



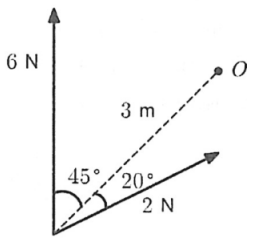
c)



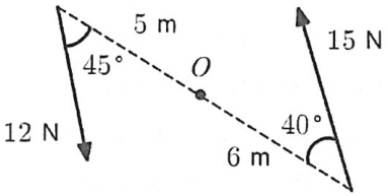
d)



e)

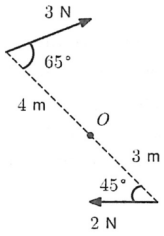


f)

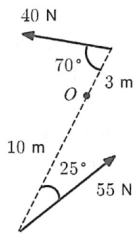


2.61 — Question 1 (continued). Find the resultant moment about O in each case, giving your answer to 3 significant figures and stating whether it is anticlockwise or clockwise:

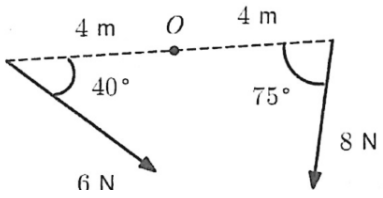
g)



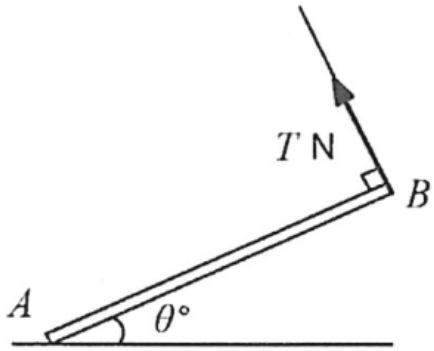
h)



i)



2.61 — Question 2. The following questions use $g = 9.8$ **ms**$^{-2}$ A uniform rod AB has mass m kg and length x metres. It rests with one end A on rough horizontal ground and is held in limiting equilibrium at angle θ by a light inextensible string at B with tension T N.



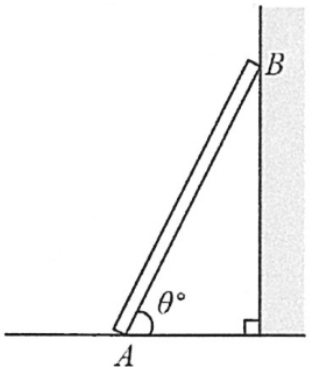
a) Draw a force diagram.

For each of the following, given

b) $m = 30, x = 10, \theta = 30$, find the value of T and the value of μ

c) $x = 8, \theta = 20, T = 180$, find the value of m and the value of μ

2.61 — Question 3. A ladder modelled as a uniform rod AB with length x metres and mass m kg rests against a wall. One end, A is resting on rough horizontal ground and the other end B rests on a smooth vertical wall. The ladder is held in limiting equilibrium at an angle of $\theta°$ to the horizontal.



The coefficient of friction between the ladder and the floor is μ

a) Draw a force diagram.

For each of the following find μ, given

b) $\theta = 60, x = 5, m = 10$

c) $\theta = 45, x = 5, m = 10$

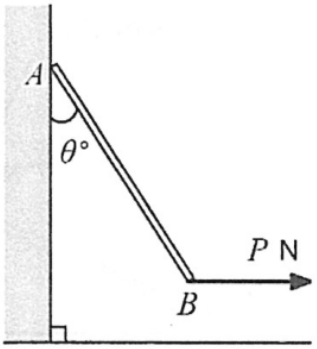
d) $\theta = 60, x = a, m = M$

A mass of $\frac{1}{2}m$ kg is loaded onto the ladder $\frac{1}{4}$ of the way up. The ladder is a metres long and has mass m kg. Find the value of μ when

e) $\theta = 60$

f) $\theta = 45$

2.61 — Question 4. A uniform lever AB of mass m kg and length x metres is attached at A to a vertical wall by a smooth hinge. The acute angle between the lever and the wall is $\theta°$. The lever is held in equilibrium by a horizontal force of magnitude P N at B.



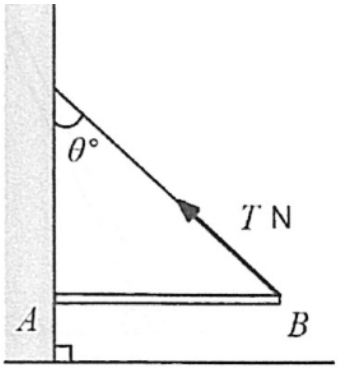
a) Draw a force diagram.

For each of the following, find P and the magnitude of the force exerted on the rod by the wall, given

b) $\theta = 30, m = 2, x = 10$

c) $\theta = 45, m = 5, x = a$

2.61 — Question 5. A uniform rod AB of mass m kg and length x metres is attached to a vertical wall by a smooth hinge. It is held in place horizontally by a light inextensible string which is attached to the wall directly above A and the other end is attached to the rod at B.



The tension in the string is T N. The acute angle between the string and the wall is $\theta°$.

a) Draw a force diagram.

For each of the following, find the value of T and the magnitude of the force exerted on the rod by the wall, given

b) $\theta = 30, m = 3, x = 10$

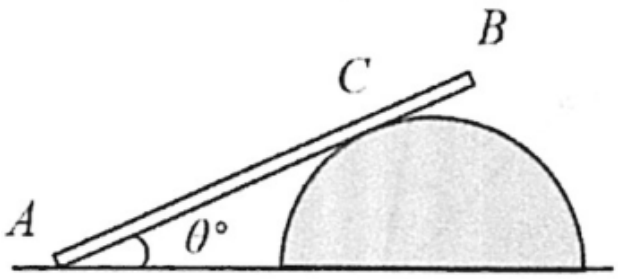
c) $\theta = 60, m = 15, x = a$

d) $\theta = 45, m = 10, x = a$

A box of mass 3 kg is added to the rod 2 metres from A. Given

e) $\theta = 40, m = 5, x = 8$, find the value of T

2.61 — Question 6. A uniform rod AB has weight 100 N and length $5a$ metres. It rests with one end A on rough horizontal ground and is resting on a smooth cylindrical drum at C where AC is $4a$ metres. The rod is angled at θ degrees to the horizontal where $\tan\theta = \frac{7}{24}$

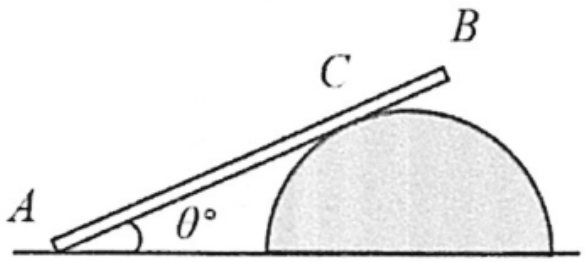


a) Draw a force diagram.

b) Find the normal and frictional components of the contact force at A.

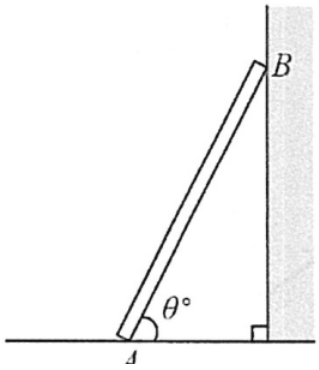
c) Find the least possible value of the coefficient of friction between the rod and the ground.

2.61 — Question 7. A non-uniform rod AB has weight 80 N which acts $2a$ metres from A. It rests with one end A on rough horizontal ground and is resting on a smooth cylindrical drum at C where AC is $4a$ metres. The rod is angled at θ degrees to the horizontal where $\tan\theta = \frac{7}{24}$



a) Draw a force diagram.
b) Find the normal and frictional components of the contact force at A.
c) Find the least possible value of the coefficient of friction between the rod and the ground.

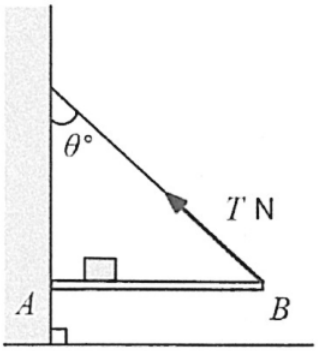
2.61 — Question 8. A uniform ladder AB of length a metres rests against a wall in limiting equilibrium. One end, A rests on rough horizontal ground at an angle of θ and the other end B rests on a smooth vertical wall. The coefficient of friction is μ



Show that

$$2\mu\tan\theta = 1$$

2.61 — Question 9. A uniform rod AB of mass 2 kg and length $4a$ metres with its end A against a rough vertical wall, with B being held in place horizontally by a light inextensible string which is attached to the wall directly above A

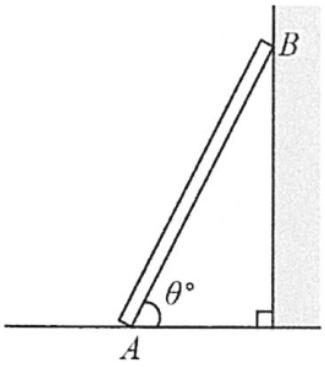


The tension in the string is T N. The acute angle between the string and the wall is $\theta°$ where $\tan\theta = \frac{3}{4}$
An object of mass 0.5 kg is placed on the rod, a metres from A

a) Find the value of T
b) Find the normal and frictional components of the contact force at A
c) Find the value of the coefficient of friction.

The following question use exact values of g

2.61 — Question 10. A ladder modelled as a uniform rod AB of length $2a$ metres and mass m kg rests against a wall. One end, A is resting on rough horizontal ground and the other end B rests on a rough vertical wall. The ladder is held in limiting equilibrium at an angle of $\theta°$ to the horizontal



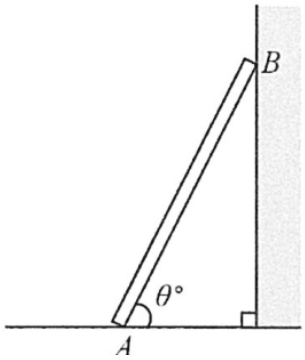
The coefficient of friction between the ladder and the floor is 0.4 and the coefficient of friction between the ladder and the wall is 0.1

a) Show that the normal reaction at the floor is 2.5 times larger than the normal reaction at the wall.

The normal reaction at the floor is 15 N, find

b) m in terms of g
c) the value of θ to 3 significant figures.

2.61 — Question 11. A uniform ladder AB of length $6a$ metres and mass m kg rests against a wall. One end, A is resting on rough horizontal ground and the other end B rests on a smooth vertical wall. The ladder is held in limiting equilibrium at an angle of $\theta°$ to the horizontal where $\tan\theta = \frac{24}{7}$



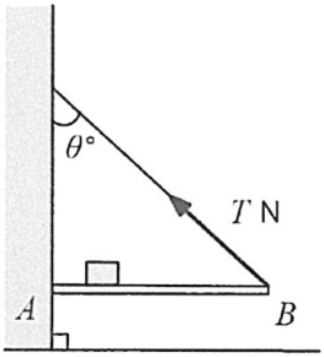
The coefficient of friction between the ladder and the floor is 0.25
A person of mass $6m$ kg stands at the top of the ladder, whilst another person holds the ladder in place, pushing with a force of magnitude P N horizontally at A.

a) Show that the reaction of the wall on the ladder is

$$\frac{91}{48}mg$$

b) Find in terms of m and g the values of P for the ladder to remain in equilibrium.

2.61 — Question 12. A uniform plank AB of mass m kg and length a metres with its end A against a rough vertical wall, with B being held in place horizontally by a light inextensible string which is attached to the wall directly above A. The acute angle between the string and the wall is $\theta°$ where $\tan\theta = \frac{4}{3}$. The tension in the string is T N.



An object of mass $5m$ kg is placed on the rod, x metres from A

a) Show that

$$T = \frac{5mg(10x + a)}{6a}$$

b) Find the % along the plank from A that the object may be placed given the maximum tension the rope can take is $8mg$ N.